AI-Based Stock Trading Strategy using Machine Learning

1. Introduction

Stock market prediction is a challenging problem due to its highly dynamic and stochastic nature. Prices are influenced by multiple factors such as economic indicators, global events, and investor sentiment.

Machine Learning provides a data-driven approach to analyze historical patterns and make predictions about future price movements.

2. Objective

The main objectives of this project are:

- To predict stock price direction (up or down)
- To develop a machine learning-based trading strategy
- To evaluate strategy performance using backtesting
- To compare model performance with market returns

3. Methodology

The project follows these steps:

Data Collection : Historical stock data is collected using the Yahoo Finance API via the yfinance library.

Feature Engineering : Several technical indicators are created:

i. Returns (percentage change)
ii. Moving averages (MA5, MA10)
iii. Volatility (rolling standard deviation)
iv. Momentum
v. RSI (Relative Strength Index)
vi. Target Variable

The target variable is defined as:

1 → if next day price increases

0 → if next day price decreases

4. Machine Learning Models

Two models are used:

i. Logistic Regression: Simple linear model, Works well for binary classification,
ii. Random Forest: Ensemble learning method, Handles non-linear relationships, Reduces overfitting

5. Data Splitting

Time-based splitting is used:

Training data: Before 2023

Testing data: After 2023

This avoids data leakage and simulates real-world prediction.

6. Model Evaluation

Evaluation metrics include:

- Accuracy
- Confusion Matrix
- Classification Report

7. Backtesting Strategy

Trading signals are generated based on prediction probabilities.

If probability > 0.6 → Buy signal

Else → No trade

Strategy returns are calculated and compared with market returns.

8. Results

- Model accuracy is approximately 50%, which is expected due to market randomness.
- The strategy shows improved risk management by reducing losses compared to the market baseline.
- Random Forest performs slightly better than Logistic Regression.

9. Concepts Covered

Supervised Learning

Classification Algorithms

Feature Engineering

Overfitting and Underfitting

Bias-Variance Tradeoff

Time Series Data Handling

10. Conclusion

The project demonstrates that Machine Learning can be applied to financial markets. While prediction accuracy is limited due to the stochastic nature of stock prices, the model helps in reducing risk and making systematic decisions.

11. Future Scope

Add more technical indicators (MACD, Bollinger Bands)

Use Deep Learning models (LSTM)

Include multiple stocks

Deploy using Streamlit